1. **Multiplicative Property**
   - If $\gcd(m, n) = 1$, then $\phi(m \cdot n) = \phi(m) \cdot \phi(n)$
2. **Closed Form Formula**
$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$
3. **Prime Power Case**
   - For prime p and $k \geq 1$:
$$\phi(p^k) = p^{k-1}(p-1) = p^k \left(1 - \frac{1}{p}\right)$$
4. **Jordan Totient Generalization**
$$J_k(n) = n^k \prod_{p|n} \left(1 - \frac{1}{p^k}\right)$$
   - $J_1(n) = \phi(n)$ counts $(k+1)$-tuples coprime with n
5. **Sum of Jordan Totients**
$$\sum_{d|n} J_k(d) = n^k$$
6. **Divisor Sum**
$$\sum_{d|n} \phi(d) = n$$
7. **Möbius Inversion Formulas**
$$\phi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d}$$
$$\phi(n) = \sum_{d|n} d \cdot \mu\left(\frac{n}{d}\right)$$
8. **Divisibility Property**
$$a \mid b \Rightarrow \phi(a) \mid \phi(b)$$
9. **Exponent Divisibility**
$$n \mid \phi(a^n - 1) \quad \text{for } a, n > 1$$
10. **General Product Formula**
$$\phi(mn) = \phi(m)\phi(n)\frac{d}{\phi(d)} \quad \text{where } d = \gcd(m, n)$$
11. **Special Cases**
    - For even numbers:
$$\phi(2m) = \begin{cases} 2\phi(m) & \text{if } m \text{ is even} \\ \phi(m) & \text{if } m \text{ is odd} \end{cases}$$

- Power formula:
$$\phi(n^m) = n^{m-1}\phi(n)$$

12. **LCM-GCD Relationship**
$$\phi(\text{lcm}(m,n)) \cdot \phi(\gcd(m,n)) = \phi(m) \cdot \phi(n)$$

13. **Parity Property**
    - $\phi(n)$ is even for $n \geq 3$
    - If n has r distinct odd prime factors, then $2^r \mid \phi(n)$

14. **Reciprocal Sum**
$$\sum_{d|n} \frac{\mu^2(d)}{\phi(d)} = \frac{n}{\phi(n)}$$

15. **Sum of Coprimes**
$$\sum_{\substack{1\leq k\leq n \\ \gcd(k,n)=1}} k = \frac{1}{2}n\phi(n) \quad \text{for } n > 1$$

16. **Radical Property**
$$\frac{\phi(n)}{n} = \frac{\phi(\text{rad}(n))}{\text{rad}(n)}$$
    - where $\text{rad}(n) = \prod_{p|n} p$

17. **Bounds**
    - Lower bound: $\phi(m) \geq \log_2 m$
    - Iterated totient bound: $\phi(\phi(m)) \leq \frac{m}{2}$

18. **Exponent Reduction**
    - For $x \geq \log_2 m$:
$$n^x \mod m = n^{\phi(m)+x \mod \phi(m)} \mod m$$

19. **GCD Sum**
$$\sum_{\substack{1\leq k\leq n \\ \gcd(k,n)=1}} \gcd(k-1,n) = \phi(n)d(n)$$
    - Also holds for $\gcd(a \cdot k - 1, n)$ when $\gcd(a,n) = 1$

20. **Non-uniqueness**
    - For every n there exists $m \neq n$ with $\phi(m) = \phi(n)$

21. **Weighted Sum**
$$\sum_{i=1}^{n} \phi(i) \left\lfloor \frac{n}{i} \right\rfloor = \frac{n(n+1)}{2}$$

22. **Odd-indexed Sum**

$$\sum_{\substack{i=1 \\ i \text{ odd}}}^{n} \phi(i) \left\lfloor \frac{n}{i} \right\rfloor = \sum_{k \geq 1} \left[\frac{n}{2^k} \right]^2$$

- where $[\cdot]$ denotes rounding

23. **Double Sum**

$$\sum_{i=1}^{n} \sum_{j=1}^{n} ij[\gcd(i,j) = 1] = \sum_{i=1}^{n} \phi(i) i^2$$

24. **Average Value**

- The average of coprimes of n less than n is $\dfrac{n}{2}$